$g(x) = (x+\alpha)(x+\alpha)(x+\alpha)(x+\alpha)(x+\alpha)(x+\alpha) =$

$= x^6 + \alpha^{10} \cdot x^5 + \ldots$

Домашнее задание 11.

Задача 1

$n = 15,\ d = 5,\ GF(2^4)$

$g(x) = (x-\alpha)(x-\alpha^2)(x-\alpha^3)(x-\alpha^4) = (x^2 - \alpha^2 x -$

$\alpha x + \alpha^3)(\alpha^2 - \alpha^4 x - \alpha^3 x + \alpha^7) = (x^2 - \alpha^5 x + \alpha^3) \cdot$

$\cdot (x^2 - \alpha^7 x + \alpha^7) = x^4 - \alpha^7 x^3 + \alpha^7 x^2 - \alpha^5 x^3 + \alpha^{12} x^2 -$

$- \alpha^{12} x + \alpha^3 x^2 - \alpha^{10} x + \alpha^{10} = x^4 - \alpha^{13} x^3 + \alpha^6 x^2 + \alpha^3 x^2 +$

$+ \alpha^3 x + \alpha^{10}$

$g(x) = x^4 + \alpha^{13} x^3 + \alpha^6 x^2 + \alpha^3 x + \alpha^{10}$

Задача 2

$PC(15,9)\ GF(2^4)\ d = 7$

$$P(x) = 1 + x + x^4$$

$$V(x) = \alpha + \alpha^3 x + \alpha^{10} x^2 + \alpha^{12} x^3 + \alpha^{13} x^4 +$$
$$+ \alpha^3 x^5 + \alpha^9 x^6 + \alpha^5 x^7 + x^8 + \alpha^{10} x^9 + \alpha^8 x^{10} +$$
$$+ \alpha x^{11} + \alpha x^{12} + \alpha^{13} x^{13} + \alpha^2 x^{14}$$

$$V = (\alpha, \alpha^3, \alpha^{10}, \alpha^{12}, \alpha^{13}, \alpha^3, \alpha^9, \alpha^5, 1, \alpha^{10}, \alpha^8, \alpha, \alpha, \alpha^{13}, \alpha^2)$$

$$S_i = \sum_{j=0}^{14} V_j (\alpha^i)^j$$

$$S_1 = \alpha + \alpha^3 \alpha + \alpha^{10} \cdot \alpha^2 + \alpha^{12} \cdot \alpha^3 + \alpha^{13} \cdot \alpha^4 + \alpha^3 \cdot \alpha^5 +$$
$$+ \alpha^9 \alpha^6 + \alpha^5 \alpha^7 + \alpha^8 + \alpha^{10} \cdot \alpha^9 + \alpha^8 \cdot \alpha^{10} + \alpha \cdot \alpha^{11} +$$
$$+ \alpha \cdot \alpha^{12} + \alpha^{13} \cdot \alpha^{13} + \alpha^2 \cdot \alpha^{14} =$$
$$= \cancel{\alpha^1} + \cancel{\alpha^4} + \cancel{\alpha^{12}} + \cancel{\alpha^0} + \alpha^2 + \cancel{\alpha^8} + \cancel{\alpha^0} + \alpha^{12} + \cancel{\alpha^8} +$$
$$+ \cancel{\alpha^4} + \alpha^3 + \cancel{\alpha^{12}} + \alpha^{13} + \alpha^{11} + \cancel{\alpha^1} = \cancel{x^2} + \cancel{1} + \cancel{x} + \cancel{x^2} + \cancel{x^3} +$$
$$+ \cancel{x^3} + \cancel{1} + \cancel{x^2} + \cancel{x^3} + \cancel{x} + \cancel{x^2} + \cancel{x^3} = \varnothing$$

аналогично:

$$S_2 = S_3 = S_4 = S_5 = S_6 = \varnothing$$

$$\Downarrow$$

принятое слово не содержит ошибок

$$e(x) = \varnothing$$